A simple rule for the evolution of cooperation on graphs and social networks

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I. PAPER IMPLEMENTATION

A. Introduction

When studying systems involving the interaction of several individuals, an aspect fundamental for their understanding is the way cooperation [2] is structured and operated. Cooperative interactions can be seen in many levels of biological organizations and from single cells to group of animals, and has been subject of much research, both theoretical and experimental, such as in behavioral economics, the prisoner's dilemma or the Nash equilibrium.

Depending on the design and the rules of the system the individuals evolve, selection (either natural or cultural) can encourage cooperating or defecting behavior. [3] For example, in unstructured populations, natural selection generally ends up being in favour of defectors over cooperators. However, in structured populations with spatial restrictions, factors to determine the overall promoted behavior are still the subject of extensive research.

The paper is putting much effort into finding a global and rather simple rule that would help determining which behavior will be promoted in systems defined by different spatial relationships or social networks. It seems that selection favours cooperation, in cases where the ratio between the altruistic act and its cost exceeds the average number of neighbours per individuals. It shows that wide-spread cooperation can be the end result of social viscosity, even in the absence of reputation effects or strategic complexity.

B. Model description

1. The Game

We consider a Prisoner's Dilemma game [1]. A cooperator pays a cost c for his opponent to receive a benefit b. A defector does not contribute anything to the other players and pays nothing.
The payoff matrix of this game is given by :

	C	D
C	b-c	-c
D	b	0

2. The graph

In the model we will reproduce, individuals can choose between two strategies. The cooperators, often noted C, pay a cost, noted c, for another individual to receive a benefit, noted b. A defector, noted D, does not contribute and hence does not pay any cost. The point here is to use different dynamics and parameters to see which cases ends up turning in favors of one situation.
The model is represented by a a graph where the individuals are the vertices and the edges denote the links between individuals in terms of interactions. Such structure can be seen in figure 1. In the model we are implementing, a sparse matrix is created, where the element ij is either 0 or 1 depending on the existence of a connection between the vertex i and the vertex j (the graph is not directed so the element ij will be equal to the element ji).
We will consider five different kinds of graphs: Cycles, where every vertex will be connected to the k/2 closest neighbours at its right and its k/2 neighbours at its left; lattices, consisting on k $= 3, 4, 6, 8$ neighbours (depending if the

lattice consist on hexagons, squares, equilateral triangles or right triangles respectively); random regular graphs, where every vertex is connected to k random neighbours; random graphs, where links are also assigned randomly but the average of links per vertex is k; and scale-free networks, where the graph is created through preferential attachment. We also considered as an initial extension the introduction of Small World networks.

3. The dynamics

We will consider a population of N individuals consisting of cooperators and defectors, where cooperators help all individuals to whom they are connected. Then, if k is the number of individuals that a cooperator is connected to and i the amount of those who are cooperators, then its payoff:

$$\mathrm{P_C} = b \cdot i - c \cdot k \tag{1}$$

Respectively, in the case of a defector we would have:

$$\mathrm{P_D} = b \cdot i \tag{2}$$

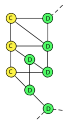


Figure 1: Graph representation of the stands within the population

In order to implement later some sort of rules, we need a way to measure the efficiency, or survivability, of a defector or cooperator stand within the global setting. We will define this term, as the fitness of an individual which will depend on the global fitness of the population, here defined as a constant, plus the payoff corresponding to this specific stand. That is in terms of an equation:

$$\mathrm{F_{individual}} = 1 - w + w \cdot \mathrm{P_{individual}} \tag{3}$$

To reach scenarios that would show social viscosity, w is preferred to be small, that is to say $w << 1$.
With the fitness as a measure, we can implement different rules regarding the evolution of the vertices. The one we will implement is a death-birth process, where one individual is uniformly randomly killed.
To do so, the information of the cooperators and defectors connected to the chosen vertex is stored, and a copy of the sparse matrix is made where all the links to the chosen vertex are erased. With this, the fitness $\mathrm{F}_{individual}$ is calculated for each neighbour through the temporal matrix (thus excluding the influence of the dead vertex, otherwise we would be considering the imitation process).
Then, the empty vertex is replaced by a cooperator or defector with probabilities:

$$\mathrm{P(C)} = \frac{\sum \mathrm{F_C}}{\sum \mathrm{F_C} + \sum \mathrm{F_D}}, \tag{4}$$

$$\mathrm{P(D)} = 1 - \mathrm{P(C)} \tag{5}$$

with $\mathrm{F_C}$ and $\mathrm{F_D}$ being respectively the fitness of a cooperator and defector stand in this vertex. This process is repeated iteratively until we observe that only one behaviour is present in the graph. Such an updating process can be seen for a cycle in figure 2. for a lattice in figure 3, and for a random regular graph in figure 4.

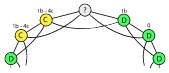


Figure 2: Updating process of a cycle with k=4.

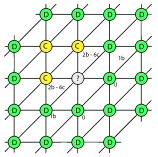


Figure 3: Updating process of a lattice with k=6.

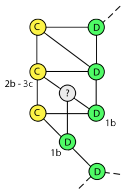


Figure 4: Updating process of a random regular graph with k=3.

4. *The fixation probability ρ*

In order to analyze the influence of the variables c, b, and k in the evolution of the model, the fixation probability measure ρ is studied. To compute ρ, a number of simulations is ran, where at the start either only one vertex is labeled as a cooperator or one vertex is labeled as a defector, and it is ran until only one behaviour prevails. Then, ρ is defined as the percentage of runs that end with all of the vertex being cooperators over the total number of runs.

If $\rho > 1/\mathrm{N}$, the choice of b,c and k will favour cooperators to become the prevailing behaviour, and conversely, if $\rho < 1/\mathrm{N}$, defectors' behaviour will be favoured.

C. Mean-field approximation

Consider the function ρ_c, the amount of cooperators a population of n vertices where every vertex has k neighbours, as a discrete function depending on t, then on each step there are two situations that can affect the number of cooperators: a cooperator being chosen to die and being replaced with a defector, or a defector being chosen to die and being replaced with a cooperator. Thus on each step we have

$$\begin{aligned}\rho_c(t+1) = \rho_c(t) - P(chos = C)P(C \to D) \\ +P(chos = D)P(D \to C)\end{aligned} \tag{6}$$

Since the dying vertex is chosen uniformly, the probability of a cooperator being chosen is $\rho_c(t)/n$, and the probability of a defector being chosen is $(n - \rho_c(t))/n$. Now,

$$P(C \to D) = \frac{F_{cd}}{F_{cc} + F_{cd}} \tag{7}$$

$$P(D \to C) = \frac{F_{dc}}{F_{dc} + F_{dd}} \tag{8}$$

where F_{cc} is the sum of the fitness of the cooperating neighbours and F_{cd} is the sum of the fitness of the defecting neighbours, given that the chosen vertex is a cooperator, and F_{dc}, F_{dd} are defined in a similar way if the chosen vertex is a defector.

The next step is to find the average cooperating and defecting neighbours of the chosen vertex. If it is a cooperator, the probability distribution of the amount of cooperating neighbours is a hypergeometric distribution $Hyp(n, \rho_c(t)-1, k)$, and its mean is $k\ (\rho_c(t)-1)/n$, and the defecting neighbours follow a $Hyp(n, n - \rho_c(t), k)$ distribution with mean $k\ (n - \rho_c(t))/n$. In the same fashion the mean of cooperating neighbours of a chosen defector is $k\ \rho_c(t)/n$, and the mean of defecting neighbours is $k\ (n-\rho_c(t)-1)/n$. Thus, we have that

$$\begin{aligned}F_{cc} &= k\frac{\rho_c(t)-1}{n}(1-\omega+\omega P_{cc}) \\ F_{cd} &= k\frac{n-\rho_c(t)}{n}(1-\omega+\omega P_{cd}) \\ F_{dc} &= k\frac{\rho_c(t)}{n}(1-\omega+\omega P_{dc}) \\ F_{dd} &= k\frac{n-\rho_c(t)-1}{n}(1-\omega+\omega P_{dd})\end{aligned} \tag{9}$$

where P_{cc} is the average fitness of a cooperating neighbour of the chosen cooperator, and P_{cd}, P_{dc}, P_{dd} are again defined in a similar way. Following the same argument that the average of cooperating and defecting neighbours follow hypergeometric distributions we get that

$$\begin{aligned}P_{cc} &= b(k-1)\big(\frac{\rho_c(t)-2}{n}+1\big) - ck \\ P_{cd} &= b(k-1)\big(\frac{\rho_c(t)-1}{n}+1\big) \\ P_{dc} &= b(k-1)\frac{\rho_c(t)-1}{n} - ck \\ P_{dd} &= b(k-1)\frac{\rho_c(t)}{n}\end{aligned} \tag{10}$$

where c and b are the cost and benefit of cooperating. Thus, we can get the ODE

$$\dot{\rho}_c = \rho_c - \frac{\rho_c}{n}\left(\frac{F_{cd}}{F_{cc}+F_{cd}}\right) + \frac{n-\rho_c}{n}\left(\frac{F_{dc}}{F_{dc}+F_{dd}}\right)$$

with $F_{cc}, F_{cd}, F_{dc}, F_{dd}$ and $P_{cc}, P_{cd}, P_{dc}, P_{dd}$ defined as above.

D. Simulation results

In our implementation, we replicated simulations for $N = 100$ for the same diversity of graph that is present in the paper[5]. The point is to be able to observe if natural selection occurs through the death birth process and favor some graphs over others by turning the whole population in cooperators. By running simulations over many randomly generated graphs of a selected type and looking at the outcome, we are able to get a fixation probability, noted ρ. In order to favor cooperation, this probability needs to be over $1/N$, which is 0.01 for us here.
We made so that the iteration process would loop until one of the outcomes would occur: only cooperators or only defectors. We generated a multitude of graph of the same type and simulated the process for each generated graph $50 \cdot N$ times to make a good compromise with bias reduction and computation time. We simulated the process on cycle [4], lattice, random regular, random graph, scale free and small world networks for a different amount of neighbors ranging from 2 to 10. Then, for each type of graph
We see in figure 5 that in most cases, if we have the ratio $b/c > k$ then natural selection favors cooperation. This is particularly the case for the cycle and lattice graphs, for which all the fixation probability nearly exactly crosses the threshold for the corresponding value of k. Though, we see that this becomes more of a requirement than a condition for higher numbers of neighbors for more complex graphs. For the random and random regular graphs[7], we see that for up to $k = 6$ the rule holds nearly perfectly, but then we need a quite higher benefit-cost ratio to reach the threshold. As the graph becomes more interconnected, the ratio requirement becomes higher. Indeed, we can see in particular that for scale free and small world networks the $b/c > k$ holds for small k values[8], but in order to pass the threshold for 10 neighbors, one need a benefit-cost ratio much higher than 10. Such a graph can be seen in figure 6. This type of graph is generally much more interconnected[6] than the more regular Cartesian graphs, therefore this explains why one would need a much higher benefit-cost ratio than the other ones.

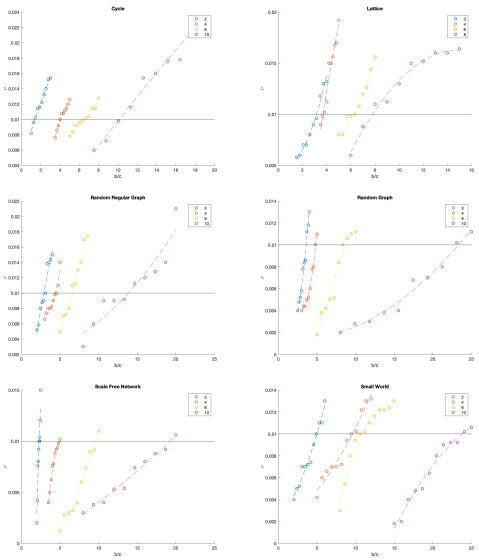


Figure 5: Benefit-cost ratio for different graphs and for various amount of neighbors

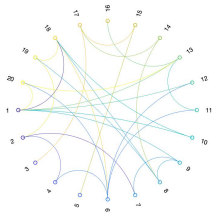


Figure 6: Small World graph

II. EXTENSION

A. Extension: Model Description

We consider the same game as in the previous section. That is the PDG with the following payoff matrix:

	C	D
C	b-c	-c
D	b	0

In the extension we consider networks with adaptive weights. That is, individuals update the strength of their connections to their neighbours according to the payoffs they receive.
In this new setting the payoff of an individual, denoted by u^i, is calculated as a weighted average taking into account its connection weights. That is,

$$u^i = \frac{1}{v^i}\sum_{j=1}^{n} w^{i,j} a_{s^i s^j} \tag{11}$$

Where $v^i = \sum_j w^{ji}$ denotes the *vertex strength* and serves as a normalizing factor. The term $a_{s^i s^j}$ refers to the payoff an individual following strategy s^i receives when interacting with an individual strategy s^j. For example, $a_{CD} = -c$ and $a_{CC} = b - c$.
On each time step a certain portion of the population undergoes a strategy update then each individual undergoes a connection weight update. The schemes for these two updates ares described in the sequel.

1. Strategy Update

We again follow the Death-Birth scheme for strategy updating [5] [10] [11]. But introduce one important parameter, β_S the rate of which individuals update their strategy.
The algorithm goes as follows: first we select $\beta_S \Delta t n$ individuals from the population at random. We apply synchronously the following two step update to the selected vertices:

1. Locate selected vertex and remove it from the graph.
2.Resulting empty vertex adopts the strategy of a neighbour j with probability proportional to u^j (success of the strategy) and w^{ij} (strength of relation between individuals). The probability of i adopting strategy s^j is denoted by $\Pi_{i\to j}$ and given by

$$\Pi_{i\to j} = \frac{w^{ij}}{u^j}\sum_{k=1}^{n} w^{ik} u^k \tag{12}$$

2. Weight Updating

We follow the weight updating scheme proposed by Ohtsuki et al 2011 [11]. In this scheme players update their neighbouring connections proportionally to the payoff they receive when interacting with each neighbour. The updating is made synchronously.
The weight w^{ij} is updating according to the following rule

$$w^{ij} \to w^{ij} + (\Delta w^{ij} + \Delta w^{ji}) B_w \Delta t \tag{13}$$

where :

$$\Delta w^{ij} = \frac{1}{\bar{u}}(a_{s^i s^j} - u^i) w^{ij} \tag{14}$$

We consider that the players i, j strengthen their connection according to their combined intentions hence why the updated term contains terms Δw^{ij} and Δw^{ji}. The term B_w is used to control the rate at which players update their weights.
We note that the graph stays symmetric, since equation (13) is symmetric with respect to i and j. Moreover:

$$\begin{aligned} \sum_j \Delta w^{ij} &= \frac{1}{\bar{u}}\sum_j a_{s^i s^j} w^{ij} \\ -\frac{1}{v^i}\sum_j a_{s^i s^j} w^{ij} &\sum_j w^{ij} = 0 \end{aligned} \tag{15}$$

Hence the sum of all weights, $\sum_i v^i$ remains a constant in the model. This is in accordance with the assumption that any individual has a fixed amount of energy and time he can invest in social relations. This energy will be allocated to the neighbours that provide most benefit.

B. Mean Field approximation

In this section we derive analytic results about the dynamics of the strength-weighted frequency of players following strategy S (denoted by Θ_S) by a mean field approximation.

1. *Case for $\delta << 1$*

We recall that $\delta = \frac{\beta_W}{\beta_S}$. Hence, in this regime of δ the rate of change of the strategy is much greater than the rate of change of the network weights. Under the mean field approximation we can assume fixed weights. We want to derive the quantity :

$$p(v,s,t) = \text{"fraction of individuals with strength } v \text{ and following strategy } s \text{ at time } t\text{"} \tag{16}$$

We work under link-dependant selection, that is a network link is selected proportional to its weight. So individuals are selected proportionally to its strength. Hence, the probability to come across strategy s while randomly selecting an individual is:

$$\Theta_S = \frac{\int_0^\infty vp(v,s)dv}{\int_0^\infty vp(v)dv} \tag{17}$$

where $p(v) = \sum_S p(v,s)$. Then on a well mixed population the average payoff of an $S-$strategist is given by :

$$u_S = \sum_{s' \in S} a_{ss'}\Theta_{S'} \tag{18}$$

In the model and under D-B updating after a vertex is emptied a new strategy is chosen proportionally to its payoff and connection weights. Under our approximations we have:

$$P(\text{"adopting strategy S"}) = \frac{u_S\Theta_S}{\bar{u}} \tag{19}$$

Where $\bar{u} = \sum_S u_s\Theta_S$ is the average payoff over all strategies. So we have:

$$p(v,s,t+\Delta t) = (1-\beta_S\Delta t)p(v,s,t) + (p(v,t)\beta_S\Delta t)(\frac{u_S\Theta_S}{\bar{u}}) \tag{20}$$

The first term represents the individuals that are not updated, the second term is the expected value of individuals changing to strategy S. Taking the limit as $\Delta t \to 0$ yields

$$\dot{p}(v,s,t) = \beta_S(-p(v,s) + p(v)\frac{u_S\Theta_S}{\bar{u}}) \tag{21}$$

So from equation (17) we derive

$$\dot{\Theta_S} = \frac{\int_0^\infty v\dot{p}(v,s)dv}{\int_0^\infty vp(v)dv} = \frac{\beta_S\int_0^\infty[-vp(v,s) + vp(v)\frac{u_s\Theta_s}{\bar{u}}]dv}{\int_0^\infty vp(v)dv} = \frac{\beta_s}{\bar{u}}(u_s - \bar{u})\Theta_s \tag{22}$$

We note that equation (22) is the form of the replicator [9] equation derived in [13], under the assumptions of a well-mixed population. Hence, it seems that the mean-filed approximation proves that it does not matter if we are in a static network or in a well-mixed population because it is essentially the same. Since an static neighbour defectors always have a bigger mean payoff than cooperators, the mean field equation under this approximations predicts that detractors will eventually wipe out cooperators on a static network. This is in direct contradiction with our findings in the previous section and with a great body of literature that agrees that cooperation can flourish in social networks. [5] [14] [15]. This difference occurs because the mean field approximation doesn't take into account the formation of stable clusters of cooperators which promote the invasion of cooperation trough the network.

2. *Case of $\delta >> 1$*

In this case we consider the strategies as fixed and we only consider changes on connection weights.

We are interested in deriving the following quantity:

$$w_{ss'} = \sum_{i,j}\sum_{s^i=s,s^j=s'} w^{ij} \tag{23}$$

That is the sum of weights w^{ij} for all the interacting individuals i and j with strategies s and s' respectively. The change of $w_{ss'}$ in one step is given by:

$$w_{ss'} \to w_{ss'} + \sum_{i,j=1}^{n} \sum_{s^i=s, s^j=s'} (\frac{1}{\bar{u}}(a_{ss'} - u^i w_{ij}) + \\ + \frac{1}{\bar{u}}(a_{s's} - u^j) w_{ji}) \beta_W \Delta t \tag{24}$$

Substituting the average payoffs of those who follow strategy s and strategy s' respectively yields:

$$\dot{w}_{ss'} = \beta_W (\frac{1}{\bar{u}}(a_{ss'} - u_s) w_{ss'} + \\ \frac{1}{\bar{u}}(a_{s's} - u_{s'}) w_{s's}) \tag{25}$$

Under the mean - field approximation the probability of an s-strategist being at one end of the link and a s'-strategist on the other end, is given by $\Theta_s \Theta'_s$. Under weight-dependant selection this probability can also be calculated directly as:

$$\frac{w_s s'}{\sum_{s,s' \in S} w_{ss'}} \tag{26}$$

Where S is the set of all possible strategies (In our case $S = \{C, D\}$). Thus, $w_{ss'}$ can be approximated as :

$$w_{ss'} \approx w_{\text{total}} \Theta_s \Theta_{s'} \tag{27}$$

where $w_{\text{total}} = \sum_{S,S' \in S} w_{ss'} = \sum_{i,j=1}^{n} w^{ij}$ is the sum of weights of all links. Hence equation (25) can be expressed as:

$$\frac{d}{dt}(w_{total} \Theta_s \Theta_{s'}) = \beta_W (\frac{1}{\bar{u}}(a_{ss'} - u_s) w_{total} \Theta_s \Theta_{s'} + \\ \frac{1}{\bar{u}}(a_{s's} - u_{s'}) w_{total} \Theta_s \Theta_{s'}) \tag{28}$$

Rearranging and cancelling terms and recalling that $u_s = \sum_{s'} a_s s' \Theta'_s$ yields

$$\dot{\Theta}_s = \frac{\beta_w}{\bar{u}} (\sum_{s'} a_{ss'} \Theta_{s'} - \bar{u}) \Theta_s \tag{29}$$

We note that the term $\sum_{s'} a_{ss'} \Theta_{s'}$ corresponds to the average payoff of the opponents of the players following strategy S, hence it corresponds to the game with transposed payoff matrix. In the sequel we use the notation $u_s^T = \sum_{s'} a_{ss'} \Theta_{s'}$ to refer to the average payoffs under the transposed payoff matrix. With this considerations our result can be expressed as

$$\dot{\Theta}_s = \frac{\beta_w}{\bar{u}} (u_s^T - \bar{u}^T) \Theta_s \tag{30}$$

From the above equation, we conclude that the strength - weighted frequency of strategy s, Θ_s, increases if $u_s^T > \bar{u}^T$, where u_s^T is the average payoff that opponents receive from s. In other words, a cooperator because he benefits the others, he gains node strength , whereas the defector gradually loses it.

3. Intermediate δ

If reproduction takes place in continuous time ($\Delta t \to$) the terms dominating dynamics of Θ are the ones derived in the previous section. Hence, from the calculations in the above cases and after rearranging we obtain :

$$\dot{\Theta}_s = + \frac{\beta_s}{\bar{u}} (u_s - \bar{u}) \Theta_s + \frac{\beta_w}{\bar{u}} (u_s^T - \bar{u}^T) \Theta_s \tag{31}$$

Which can be expressed as:

$$\dot{\Theta}_s = \frac{1}{\bar{u}} (\sum_{s'} (\beta_S \alpha_{ss'} + \beta_W \alpha_{s's}) \Theta_{s'} \\ - (\beta_S + \beta_W) \bar{u}) \Theta_s \tag{32}$$

This is equivalent to the replicator equation with the payoff matrix, $\beta_S A + \beta_W A^T$. Hence, the payoff matrix is directly determined by the balance between the update rate of strategies and update rate of the weights. In our case:

$$\beta_S A + \beta_W A^T = \\ \begin{pmatrix} (b\beta_S - c\beta_W) + (b\beta_W - c\beta_S) & b\beta_W - c\beta_S \\ b\beta_S - c\beta_W & 0 \end{pmatrix} \tag{33}$$

Hence we predict that if $\frac{b-c\delta}{c-b\delta} > k$ selection favours cooperators.
Moreover we see that

$$\frac{d}{d\delta}(\frac{b-c\delta}{c-b\delta}) = \frac{b^2-c^2}{(c-b\delta)^2} > 0 \qquad (34)$$

Provided that $b > c$, hence we predict that increases in δ will produce increases in cooperation. Also we note that for somewhat bigish δ the term $(c-b\delta)^2$ becomes big. Hence the effect on cooperation for big values of delta is rapidly atenuated.

C. Numerical Simulations

We focused on studying the evolution of cooperators in four different graphs: Random graph (RG) , regular random graph (RRG), Scale Free Network (SFN) and small world netwrok (SWN). We study the behaviour of the system for different values of the ratio $\frac{b}{c}$ and δ, in ranges of $\delta << 1$, $\delta \sim 1$, $\delta >> 1$. During this section we switch to the measure $p_C(T) =$ "fraction of cooperators after T time steps" . The initial strategy of each vertex is initialized at random and with equal probability between defection or cooperation. A strategy will be favoured if $p_C(T)$ increases on average over a determined number of independent simulations. [18]. The graph weights are initialized at 1.

1. *Simulations for small values of δ*

For very small values of $\delta = \frac{B_w}{B_s}$ the rate at which individuals change strategy is some orders of magnitude bigger that the rate at which individuals update their connection weights, hence it is as if we were working with static weights. The results should mimic the model of the previous section, so we should see an increase of cooperation when increasing the cost-benefit ratio and that selection favours cooperators for $\frac{b}{c} > 4$. This cannot be seen in our figure for $\delta = 0.001$ (or for any fixed value of δ). Possibly this is due to our measure of choice not being precise

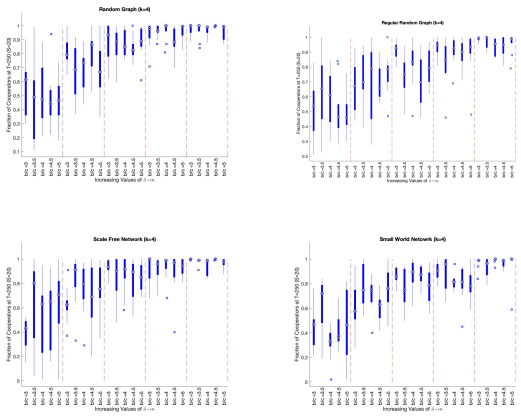


Figure 7: Figure showing the distribution of final fraction of cooperators after $T = 250$ steps over 20 independent runs. The values are obtained for $\frac{b}{c} = 3, 3.5, 4, 4.5, 5$ and $\delta = (0.001, 0.05, 0.1, 0.15, 0.25)$ where each section in between dotted red lines corresponds to a value of δ starting from $\delta = 0.001$.

enough and too susceptible to the intrinsic randomness in the model. However, what we can clearly appreciate is an increase of cooperation as δ grows. This is in accordance with our derivations of section II B 3. When the update on weights has a sizable effect detractors see their connections to their neighbours, and thus their payoff, rapidly decreasing. Cooperators on the other hand enjoy stronger connections to their cooperating neighbours and thus see their payoff increased. For $\delta = 0.15, 0.25$ cooperation rate goes above 0.8 in (almost) all simulations after only 250 steps. This stays true for all studied graphs.

We can further plot phase transition diagrams for different values of $\frac{b}{c}$, our network structure of choice is a Small World network with $p = 0.05$ for its close relation to real-world human social networks [16] [17]. This is shown in Figure 8

Again we can see that δ improves the spread of cooperation. It seems that for small values of b/c cooperation is somehow preferred, this can also be observed for some of the graphs in Figure 7 which were produced on an independent simulation. We attribute this at the our measure of choice being too susceptible to random-

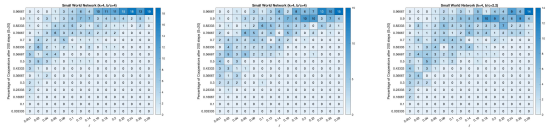


Figure 8: Phase transition diagram for a Small World network and different values of b/c.

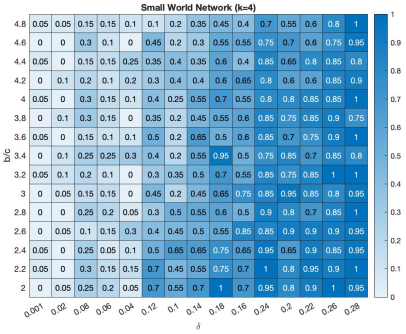


Figure 9: Fraction of realizations in which cooperators surpass the 0.85 threshold after 200 steps. Each data point is obtained by averaging over 20 simulations

ness.

Finally we can show how the parameters b/c and δ combine to affect the spread of cooperation. This is shown on Figure 9 . We see that b/c (at least for our measure of choice) is a much less important parameter than δ and increases in b/c barerly affect cooperation while increases on δ heavily benefit it.

2. *Simulations for $\delta \approx 1$*

For $0.8 < \delta < 1.2$ we found that after 250 steps the cooperation rate reaches 1 in all simulations. To see the effects of δ we can plot the evolution of cooperation over time for different values of δ.

In figure 10 we see that independently of the network structure and value of δ selection heavily benefits cooperation.

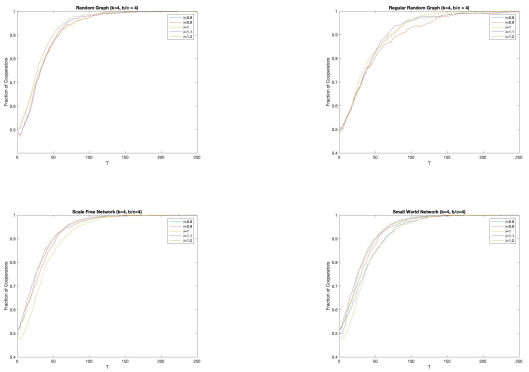


Figure 10: Figure showing the mean evolution of cooperators over time. This is plotted for for fixed value of $k = 4$ and $b/c = 4$.

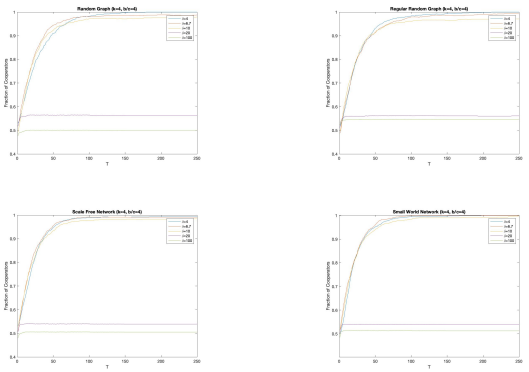


Figure 11: Figure showing the mean evolution of cooperators over time. This is plotted for for fixed value of $k = 4$ and $b/c = 4$.

3. *Simulations for big values of δ*

We plotted the mean evolution of cooperators for $\delta = 4, 6.7, 10, 20, 100$ this is shown in figure 11. We can see that the evolution over time doesn't look much different than for $\delta \sim 1$. The amount of cooperators only increases slightly faster with higher values of δ. What is interesting to note is that for $\delta = 6.7, 10$ the cooperation rate doesn't reach 1 as an equilibrium, and for $\delta = 20, \delta = 100$ the cooperation rate stabilizes a bit above 0.5. This is because individuals update their weights way too fast and detract-

ors become isolated before they can "reevaluate" their strategy. This produces an equilibrium between detractors with node strength 0 and a connected network of cooperators.

III. CONCLUSION

In the original paper, the writers found that the benefit-cost versus the number of neighbors rule, holds for a wide range of typical network structures. That is, cooperation is favoured as long as $b/c > k$. Our implementation shows that this is not always the case, especially for complex graphs and higher number of neighbors. The difference could come from the number of simulations we produced compared to the ones produced by the original authors. We ran fewer simulations and re-initialized the graphs fewer times. Thus, stochastic variation might affect our results.

However, we could still show that the rule $b/c > k$ is a remarkably good approximation to estimate when selection favours cooperation. These results hold for all the different graphs that we studied.

On the second part of our project we extended the model to include connection weight updating and studied how the ratio $\delta = \frac{\beta_w}{\beta_s}$ influenced the spread of cooperation. Our results show that the spread of cooperation rapidly increases for values of δ in the range $(0.001, 1)$. For bigger values of δ this effect is heavily attenuated and the dynamics stay very similar even with significant increases of δ. This is in accordance with the theoretical results derived from the mean-field approximation. In the range of $\delta > 6$, we showed that cooperation cannot spread to the whole graph since defectors network strength approaches zero in few time steps hence, defectors become isolated from the remaining network of cooperators. The isolated won't favour defection nor cooperation. These results hold for a variety of network structures: RG, RRG, SFN and SW.

These results could be applied in human organizations. We showed that a small punishment of individuals following a defector strategy in the form of reduced network connections (or by artificially reducing their payoff) has a dramatic effect to the promotion of cooperation. Too much punishment and defectors will become isolated and will not change their strategy while the rate of spread of cooperation will suffer no significant improvement. We conclude: Punish, but not too hard.

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[18] This was done simply because we lacked computer power to use the original measure fixation probability. The figures on this section already took more than 140 hours of computer simulations distributed among 9 computers.